# Database Project Report Template

Due Date: [Insert Date - Week ….]
Project ID & Title: (Healthcare Clinic & Telemedicine Portal)

## A. Project Identity

Team Name: G2
Team Members:


- Lê Nguyễn Quốc Hùng n24dece070@student.ptithcm.edu.vn
- Trần Hoàng Hải Đăng n24dece059@student.ptithcm.edu.vn
- Trần Quốc Huy n24dece074@student.ptithcm.edu.vn


Project Title:

## MediConnect: Integrated Healthcare Clinic & Telemedicine Management System

## B. Report structure

### 1. Introduction & Project Scope (Adapted from ISO/IEC/IEEE 29148)

The Healthcare Clinic & Telemedicine Portal is a centralized database system designed to support comprehensive healthcare information management, clinical workflow coordination, and administrative operations for a clinic providing both traditional in-person consultations and telemedicine services. The system manages multi-role user authentication and access control, patient profiles, doctor information, doctor schedules, appointment booking, follow-up appointments, consultation sessions, medical histories, and digital prescriptions.

The system is intended to keep healthcare information consistent, secure, accurate, and reliable. It also prevents scheduling conflicts and enforces strong relationships among doctors, schedules, appointments, consultation sessions, medical histories, and prescriptions.

Role-Based Access Control (RBAC) is applied with three main roles: ADMIN, DOCTOR, and PATIENT. General Practitioners and Specialists share common doctor information, so DOCTOR is modeled as a supertype with GENERAL_PRACTITIONER and SPECIALIST as total and disjoint subtypes.

#### 1.1 System Objective

The main objective of the system is to provide a reliable database for storing and managing the essential information of a healthcare clinic.

- Manage user credentials and enforce Role-Based Access Control (RBAC) across ADMIN, DOCTOR, and PATIENT roles.
- Maintain centralized patient profiles containing demographic, contact, emergency contact, and baseline clinical information.
- Manage doctor information and professional credentials.
- Distinguish General Practitioners from Specialists using Total and Disjoint EER specialization.
- Link medical specialists to their corresponding clinical specialties.

- Manage doctors' working schedules and availability while preventing overlapping schedule slots.
- Coordinate appointment bookings and support patient self-booking, administrator-assisted booking, and doctor-created follow-up appointments.
- Manage consultation sessions and enforce that each consultation session is based on a valid appointment.
- Record actual consultation start time, end time, duration, modality, and clinical notes.
- Maintain longitudinal medical history and diagnosis records.
- Manage prescriptions issued by attending doctors during valid clinical sessions.
- Maintain data integrity and prevent inconsistent healthcare records.
- Provide a rigorous conceptual database model for later logical design, SQL implementation, and Python web development.

## 1.2 Business Rules & Constraints

### 1.2.1 User Account

BR1: Each user account must have a unique UserID.

BR2: Each user account must have a unique, non-null Username and a secure password hash.

BR3: Every account must be assigned exactly one role: ADMIN, DOCTOR, or PATIENT.

BR4: Every registered doctor and patient must be linked to exactly one valid USER_ACCOUNT.

BR5: An account cannot be deleted while active dependent operational records require it.

### 1.2.2 Patient

BR6: Each patient must have a unique PatientID.

BR7: Each patient must have basic identifying and contact information.

BR8: A patient profile records baseline health information including blood type, allergies, chronic diseases, and emergency contact information.

BR9: A patient can have zero, one, or many appointments over time.

BR10: Each appointment belongs to exactly one patient.

BR11: A patient can participate in zero, one, or many consultation sessions.

BR12: A patient can have multiple medical history records and multiple prescriptions.

### 1.2.3 Doctor and Medical Staff

BR13: Each doctor must have a unique DoctorID.

BR14: Each doctor must have full name, contact information, and a unique LicenseNumber.

BR15: Every doctor must be classified as either a General Practitioner or a Specialist.

BR16: GENERAL_PRACTITIONER is a subtype of DOCTOR.

BR17: SPECIALIST is a subtype of DOCTOR.

BR18: Each Specialist must be associated with exactly one SPECIALTY.

BR19: GENERAL_PRACTITIONER and SPECIALIST are disjoint; a doctor cannot belong to both.

BR20: The specialization is total; every DOCTOR belongs to one of the two subtypes.

1.2.4 Doctor Schedule

BR21: A doctor can have zero, one, or many schedule records.

BR22: Each schedule record belongs to exactly one doctor.

BR23: Each schedule specifies ScheduleDate, StartTime, EndTime, and AvailabilityStatus.

BR24: A doctor must not have overlapping working schedule periods on the same date.

BR25: An appointment may only use an active and available working schedule.

1.2.5 Appointment

BR26: Each appointment must have a unique AppointmentID.

BR27: Each appointment must be associated with exactly one patient and exactly one doctor.

BR28: An appointment can be created by a patient, ADMIN, or assigned DOCTOR according to system permissions.

BR29: Each appointment must specify date/time, estimated duration, and modality: In-person or Telemedicine.

BR30: Status must be Scheduled, Checked-In, Completed, Cancelled, or No-show.

BR31: A doctor-created follow-up appointment must reference the preceding appointment through a recursive relationship.

BR32: If a patient does not attend, the appointment status must be updated to No-show.

BR33: Cancelled and No-show appointments remain recorded in appointment history.

BR34: A doctor must not have overlapping active appointments.

BR35: Cancelled and No-show appointments are not treated as active appointments for conflict checking.

### 1.2.6 Consultation Session

BR36: Each consultation session must have a unique SessionID.

BR37: A consultation session can only exist for a valid appointment.

BR38: Each consultation session must be linked to exactly one appointment and must use the patient and doctor assigned to that appointment.

BR39: Each session records exact start time, end time, and computed actual duration in minutes.

BR40: Each session records clinical notes and whether it is In-person or Virtual.

### 1.2.7 Medical History

BR41: A patient may accumulate multiple medical history records.

BR42: Each medical history record belongs to one patient and is linked to the consultation session in which the diagnosis was established.

BR43: A record contains diagnosis, symptoms, record date, and progress notes.

BR44: Previous medical history records are permanent clinical archives and should not be deleted.

### 1.2.8 Prescription

BR45: Each prescription must have a unique PrescriptionID.

BR46: A prescription must be issued for a valid patient by a valid doctor and linked to a valid consultation session.

BR47: The prescription doctor must be the attending doctor assigned to the related consultation and appointment.

BR48: Each prescription must contain one or more PRESCRIPTION_ITEM records.

### 1.2.9 Medication and Prescription Item

BR49: Each prescription item must reference a valid MEDICATION and define dosage, frequency, duration, and special instructions.

### 1.2.10 Main Constraints
Cross-Entity, Workflow, and Telemedicine Rules

BR50: APPOINTMENT.DoctorID must match the DoctorID of the selected DOCTOR_SCHEDULE.

BR51: AppointmentDateTime must occur on the selected ScheduleDate and fall within its StartTime-EndTime interval.

BR52: MEDICAL_HISTORY can only be created when the related APPOINTMENT has Status = Completed.

BR53: PRESCRIPTION can only be created when the related APPOINTMENT has Status = Completed.

BR54: PRESCRIPTION.PatientID must match the PatientID of the referenced APPOINTMENT.

BR55: PRESCRIPTION.DoctorID must match the DoctorID of the referenced APPOINTMENT.

BR56: The CONSULTATION_SESSION referenced by PRESCRIPTION or MEDICAL_HISTORY must belong to the same APPOINTMENT referenced by that clinical record.

BR57: A telemedicine appointment must use a Virtual consultation session when the consultation is conducted remotely.

BR58: A Virtual consultation session may store a meeting URL or virtual consultation reference required for remote patient-doctor interaction.

Integrity Constraints

- IC1 - Entity Integrity: Every primary entity must have a unique, non-null primary key.
- IC2 - Referential Integrity: Every foreign key must reference an existing parent key.
- IC3 - Domain Integrity: Role, AppointmentType, Status, AvailabilityStatus, dates, and timestamps must conform to defined domains.
- IC4 - Uniqueness Constraints: Username, PatientID, DoctorID, and LicenseNumber must be unique where required.
- IC5 - Specialization Integrity: Every doctor belongs to exactly one of the two total and disjoint subtypes.
- IC6 - Scheduling Integrity: A doctor cannot have overlapping working schedules or overlapping active appointments.
- IC7 - Workflow Integrity: A consultation session cannot exist without a valid appointment; medical history and prescriptions require a completed appointment.
- IC8 - Temporal Integrity: AppointmentDateTime must match the selected schedule date and time interval; consultation start/end times must be logically ordered.
- IC9 - Cross-Entity Consistency: The appointment doctor must own the selected schedule.
- IC10 - Clinical Record Consistency: Prescription and medical history references must remain consistent with the same appointment and consultation session.
- IC11 - Telemedicine Integrity: When an appointment is conducted as Telemedicine, the corresponding consultation session must use SessionType = Virtual.

## 2. Database Design (ISO/IEC 19505 / IE Standards)

### 2.1 Conceptual Model (ER/EER Diagram)

The conceptual model represents the main entities, attributes, relationships, cardinalities, and specialization required by the Healthcare Clinic & Telemedicine Portal.

ENTITY OVERVIEW

| Entity | Primary Key | Purpose |
|---|---|---|
| USER_ACCOUNT | UserID | Stores user login credentials and role information for ADMIN, |

| Entity | Primary Key | Purpose |
|---|---|---|
| | | DOCTOR, and PATIENT users. |
| PATIENT | PatientID | Stores patient personal, contact, and basic healthcare information. |
| DOCTOR | DoctorID | Stores common personal and professional information of doctors. |
| GENERAL_PRACTITIONER | DoctorID | Represents General Practitioners as a subtype of DOCTOR. |
| SPECIALIST | DoctorID | Represents Specialist doctors and links them to a medical specialty. |
| SPECIALTY | SpecialtyID | Stores medical specialty information. |
| DOCTOR_SCHEDULE | ScheduleID | Stores doctors' available working schedules and time periods. |
| APPOINTMENT | AppointmentID | Stores appointments between patients and doctors, including appointment type, status, and follow-up information. |
| CONSULTATION_SESSION | SessionID | Stores actual consultation sessions created from valid appointments. |
| MEDICAL_HISTORY | MedicalHistoryID | Stores patients' diagnosis, symptoms, and progress notes from consultation sessions. |
| PRESCRIPTION | PrescriptionID | Stores digital prescriptions issued during consultation sessions. |
| PRESCRIPTION_ITEM | PrescriptionItemID | Connects prescriptions with medications and stores dosage, frequency, duration, and instructions. |
| MEDICATION | MedicationID | Stores medication information used in prescriptions. |

USER_ACCOUNT and Role Structure

The system uses USER_ACCOUNT as the central entity for Role-Based Access Control (RBAC). Each account is assigned exactly one role: ADMIN, DOCTOR, or PATIENT.

Relationships: USER_ACCOUNT -> DOCTOR: 1 : 0..1; USER_ACCOUNT -> PATIENT: 1 : 0..1. Every DOCTOR and PATIENT profile must be associated with exactly one USER_ACCOUNT.

RELATIONSHIP SUMMARY

| **Relationship** | **Cardinality** | **Business Rule** |
|---|---|---|
| USER_ACCOUNT – PATIENT | 1 : 0..1 | A user account may be associated with at most one Patient profile, and each Patient must have one User Account. |
| USER_ACCOUNT – DOCTOR | 1 : 0..1 | A user account may be associated with at most one Doctor profile, and each Doctor must have one User Account. |
| PATIENT – APPOINTMENT | 1 : N | A Patient may book many Appointments, while each Appointment belongs to exactly one Patient. |
| DOCTOR – APPOINTMENT | 1 : N | A Doctor may handle many Appointments, while each Appointment is assigned to exactly one Doctor. |
| DOCTOR – DOCTOR_SCHEDULE | 1 : N | A Doctor may have many schedule records, while each Doctor Schedule belongs to one Doctor. |
| DOCTOR_SCHEDULE – APPOINTMENT | 1 : N | One Doctor Schedule may be used for multiple Appointments, while each Appointment references one Doctor Schedule. |
| DOCTOR – GENERAL_PRACTITIONER | 1 : 0..1 | A Doctor may be classified as a General Practitioner. |
| DOCTOR – SPECIALIST | 1 : 0..1 | A Doctor may be classified as a Specialist. |
| SPECIALTY – SPECIALIST | 1 : N | One Specialty may contain many Specialists, while each Specialist belongs to exactly one Specialty. |
| APPOINTMENT – APPOINTMENT | 1 : 0..N | An Appointment may be the original appointment |

| Relationship | Cardinality | Business Rule |
|---|---|---|
| | | for one or more follow-up appointments. |
| APPOINTMENT – CONSULTATION_SESSION | 1 : 0..1 | An Appointment may result in at most one Consultation Session, and each Consultation Session belongs to one Appointment. |
| APPOINTMENT – MEDICAL_HISTORY | 1 : N | An Appointment may be associated with multiple Medical History records. |
| CONSULTATION_SESSION – MEDICAL_HISTORY | 1 : N | One Consultation Session may generate multiple Medical History records. |
| CONSULTATION_SESSION – PRESCRIPTION | 1 : N | One Consultation Session may issue multiple Prescriptions. |
| PATIENT – MEDICAL_HISTORY | 1 : N | A Patient may have many Medical History records, while each record belongs to one Patient. |
| PATIENT – PRESCRIPTION | 1 : N | A Patient may receive many Prescriptions, while each Prescription belongs to one Patient. |
| DOCTOR – PRESCRIPTION | 1 : N | A Doctor may issue many Prescriptions, while each Prescription is issued by one Doctor. |
| PRESCRIPTION – PRESCRIPTION_ITEM | 1 : N | One Prescription contains one or more Prescription Items. |
| MEDICATION – PRESCRIPTION_ITEM | 1 : N | One Medication may appear in many Prescription Items, while each Prescription Item references one Medication. |

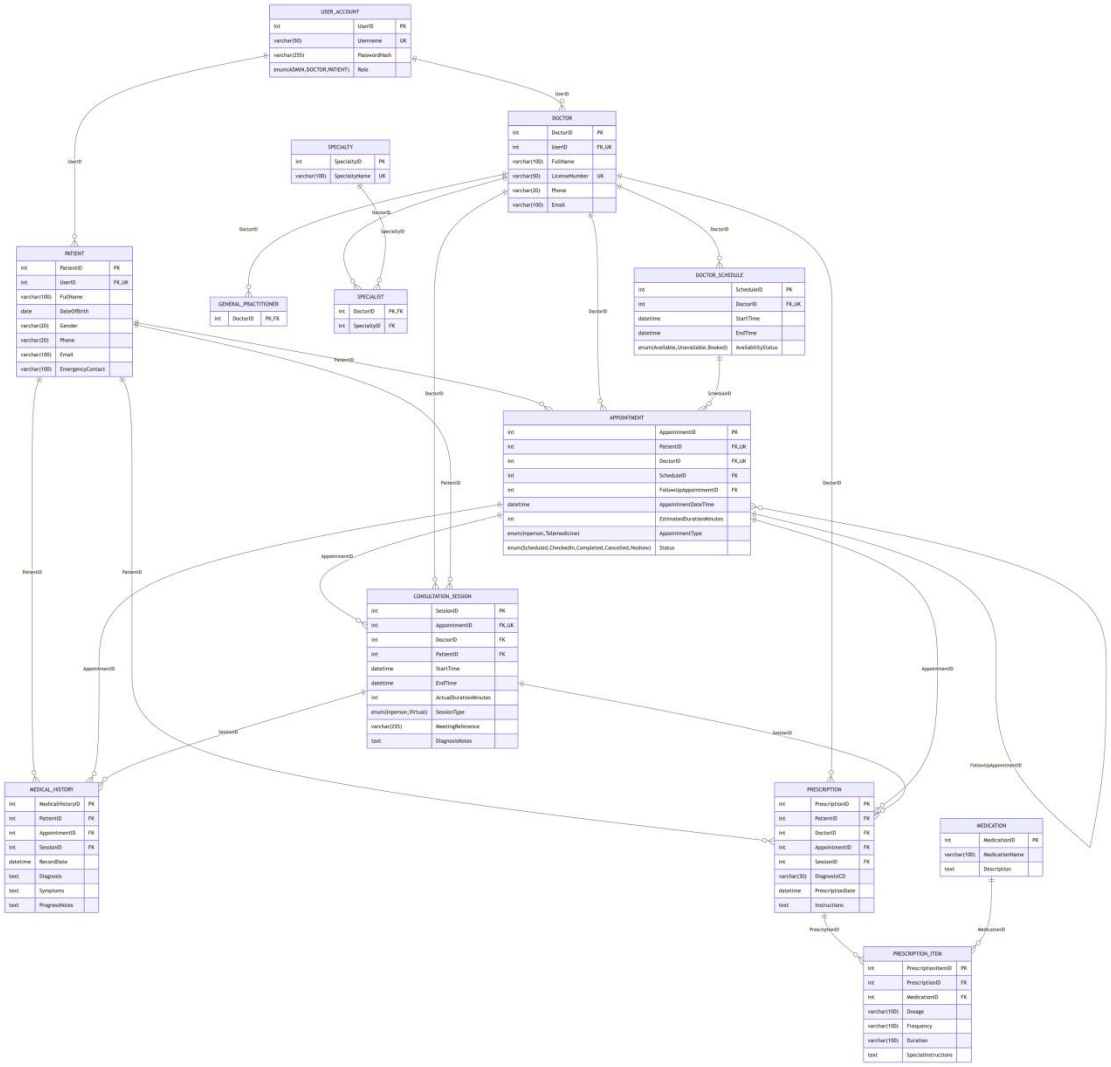
EER Specialization

DOCTOR is specialized into two subtypes: GENERAL_PRACTITIONER and SPECIALIST. SPECIALIST is associated with SPECIALTY.

The specialization is Total and Disjoint. Total specialization means every DOCTOR must belong to one subtype. Disjoint specialization means a doctor cannot simultaneously be both a GENERAL_PRACTITIONER and a SPECIALIST.

Conceptual ER/EER Diagram

The ER/EER diagram should represent the entities and relationships above, including the recursive follow-up relationship for APPOINTMENT and the Total + Disjoint specialization of DOCTOR into GENERAL_PRACTITIONER and SPECIALIST.




## 2.2 Logical Schema Mapping

The conversion from the EER diagram to a relational database schema strictly follows the standard 8-step mapping algorithm by Elmasri & Navathe:

Step 1: Mapping of Regular Entity Types

For each strong entity type E in the EER schema, create a relation R that includes all simple attributes of E. Choose one of the key attributes of E as the Primary Key (PK) of R.

- USER_ACCOUNT (UserID, Username, PasswordHash, Role)
- PATIENT (PatientID, FullName, DateOfBirth, Gender, Phone, Email, EmergencyContact)
- SPECIALTY (SpecialtyID, SpecialtyName)
- MEDICATION (MedicationID, MedicationName, Description)

Step 2: Mapping of Weak Entity Types

- In MediConnect, PRESCRIPTION_ITEM is modeled as an associative/dependent entity referencing PRESCRIPTION and MEDICATION using a surrogate primary key PrescriptionItemID alongside composite candidate keys.

Step 3: Mapping of Binary 1:1 Relationship Types

For each binary 1:1 relationship, identify the participating relations and include the primary key of one relation as a Foreign Key (FK) in the other, enforced with a UNIQUE constraint.

- USER_ACCOUNT – PATIENT (*1: 0. . 1*): Include UserID (FK, UNIQUE) in PATIENT.
- USER_ACCOUNT – DOCTOR (*1: 0. . 1*): Include UserID (FK, UNIQUE) in DOCTOR.
- APPOINTMENT – CONSULTATION_SESSION (*1: 0. . 1*): Include AppointmentID (FK, UNIQUE) in CONSULTATION_SESSION.

Step 4: Mapping of Binary 1:N Relationship Types

For each regular binary 1:N relationship, identify relation S representing the N-side and include the primary key of relation T representing the *1*-side as a foreign key in S.

- DOCTOR – DOCTOR_SCHEDULE (1:N): Include DoctorID (FK) in DOCTOR_SCHEDULE.
- PATIENT – APPOINTMENT (1:N): Include PatientID (FK) in APPOINTMENT.
- DOCTOR – APPOINTMENT (1:N): Include DoctorID (FK) in APPOINTMENT.
- DOCTOR_SCHEDULE – APPOINTMENT (1:N): Include ScheduleID (FK) in APPOINTMENT.
- SPECIALTY – SPECIALIST (1:N): Include SpecialtyID (FK) in SPECIALIST.
- CONSULTATION_SESSION – MEDICAL_HISTORY (1:N): Include SessionID (FK) in MEDICAL_HISTORY.
- CONSULTATION_SESSION – PRESCRIPTION (1:N): Include SessionID (FK) in PRESCRIPTION.
- PATIENT / DOCTOR / APPOINTMENT – PRESCRIPTION (1:N): Include PatientID (FK), DoctorID (FK), and AppointmentID (FK) in PRESCRIPTION.
- PRESCRIPTION / MEDICATION – PRESCRIPTION_ITEM (1:N): Include PrescriptionID (FK) and MedicationID (FK) in PRESCRIPTION_ITEM.

Step 5: Mapping of Recursive 1:N Relationship Types

- APPOINTMENT – APPOINTMENT (1:0..N - Follow-up): Include FollowUpAppointmentID (FK, Nullable) in APPOINTMENT referencing AppointmentID within the same relation.

Step 6: Mapping of EER Specialization (Option 8A: Superclass & Subclasses)

The superclass DOCTOR is specialized into two subtypes, GENERAL_PRACTITIONER and SPECIALIST, with Total and Disjoint constraints. Following Option 8A (Elmasri):

+ Superclass Relation: DOCTOR (DoctorID, UserID, FullName, LicenseNumber, Phone, Email) containing all shared attributes.
+ Subclass Relation 1: GENERAL_PRACTITIONER (DoctorID*) where DoctorID acts as both Primary Key and Foreign Key referencing DOCTOR(DoctorID).
+ Subclass Relation 2: SPECIALIST (DoctorID* SpecialtyID*) where DoctorID acts as both Primary Key and Foreign Key referencing DOCTOR(DoctorID), with SpecialtyID (FK) referencing SPECIALTY(SpecialtyID).

2.2.1 Relational Schema Summary

Underlined attributes represent Primary Keys (PK), while attributes marked with an asterisk (*) represent Foreign Keys (FK).

+ USER_ACCOUNT (UserID, Username, PasswordHash, Role)
+ PATIENT (PatientID, UserID*, FullName, DateOfBirth, Gender, Phone, Email, EmergencyContact)
+ DOCTOR (DoctorID, UserID*, FullName, LicenseNumber, Phone, Email)
+ GENERAL_PRACTITIONER (DoctorID*)
+ SPECIALIST (DoctorID*SpecialtyID*)
+ SPECIALTY (SpecialtyID, SpecialtyName)

+ DOCTOR_SCHEDULE (ScheduleID, DoctorID*, ScheduleDate, StartTime, EndTime, AvailabilityStatus)
+ APPOINTMENT (AppointmentID, PatientID*, DoctorID*, ScheduleID*, FollowUpAppointmentID*, AppointmentDateTime, EstimatedDurationMinutes, AppointmentType, Status)
+ CONSULTATION_SESSION (SessionID, AppointmentID*, DoctorID*, PatientID*, StartTime, EndTime, ActualDurationMinutes, SessionType, MeetingReference, DiagnosisNotes)
+ MEDICAL_HISTORY (MedicalHistoryID, PatientID*, AppointmentID*, SessionID*, RecordDate, Diagnosis, Symptoms, ProgressNotes)
+ PRESCRIPTION (PrescriptionID, PatientID*, DoctorID*, AppointmentID*, SessionID*, DiagnosisICD, PrescriptionDate, Instructions)
+ MEDICATION (MedicationID, MedicationName, Description)
+ PRESCRIPTION_ITEM (PrescriptionItemID, PrescriptionID*, MedicationID*, Dosage, Frequency, Duration, SpecialInstructions)

## 2.3 Normalization Verification (Formal proofs for 1NF → 3NF/BCNF)

This section provides formal mathematical proofs verifying that the relational schema satisfies 1NF, 2NF, 3NF, and Boyce-Codd Normal Form (BCNF).

### 2.3.1 Definition of Functional Dependencies (FDs)

Let X → Y represent a Functional Dependency where X functionally determines Y. The set of functional dependencies F for primary relations in MediConnect is defined as follows:

#### 2.3.1.1 Relation USER_ACCOUNT

- $FD_1$: UserID → {Username, PasswordHash, Role}
- $FD_2$ : Username → {UserID, PasswordHash, Role}
- Candidate Keys: {UserID}, {Username}

#### 2.3.1.2 Relation PATIENT

- $FD_1$ : PatientID → {UserID, FullName, DateOfBirth, Gender, Phone, Email, EmergencyContact}
- $FD_2$ : UserID → {PatientID, FullName, DateOfBirth, Gender, Phone, Email, EmergencyContact}
- Candidate Keys: {PatientID}, {UserID}

#### 2.3.1.3 Relation DOCTOR

- $FD_1$ : DoctorID → {UserID, FullName, LicenseNumber, Phone, Email}
- $FD_2$ : LicenseNumber → {DoctorID, UserID, FullName, Phone, Email}
- $FD_3$ : UserID → {DoctorID, FullName, LicenseNumber, Phone, Email}
- Candidate Keys: {DoctorID}, {LicenseNumber}, {UserID}

#### 2.3.1.4 Relation APPOINTMENT

- $FD_1$ : AppointmentID → {PatientID, DoctorID, ScheduleID, FollowUpAppointmentID, AppointmentDateTime, EstimatedDurationMinutes, AppointmentType, Status}
- Candidate Key: {AppointmentID}

2.3.1.5 Relation CONSULTATION_SESSION

- $FD_1$ : SessionID → {AppointmentID, DoctorID, PatientID, StartTime, EndTime, ActualDurationMinutes, SessionType, MeetingReference, DiagnosisNotes}
- $FD_2$ : AppointmentID → {SessionID, DoctorID, PatientID, StartTime, EndTime, ActualDurationMinutes, SessionType, MeetingReference, DiagnosisNotes}
- Candidate Keys: {SessionID}, {AppointmentID}

2.3.1.6 Relation PRESCRIPTION_ITEM

- $FD_1$ : PrescriptionItemID → {PrescriptionID, MedicationID, Dosage, Frequency, Duration, SpecialInstructions}
- $FD_2$ : {PrescriptionID, MedicationID} → {PrescriptionItemID, Dosage, Frequency, Duration, SpecialInstructions}
- Candidate Keys: {PrescriptionItemID}, {PrescriptionID, MedicationID}

2.3.2 First Normal Form (1NF) Proof

Definition: A relation R is in 1NF if and only if all attribute values are atomic and there are no repeating groups or composite/multivalued attributes.

Proof:

- Attribute Atomicity: All scalar fields (AppointmentDateTime, StartTime, EndTime, LicenseNumber, Role) use atomic data types (INT, VARCHAR, DATETIME, ENUM). No field contains arrays, JSON blocks, or delimited strings.
- Absence of Repeating Groups:
    - Medication entries in prescriptions are separated into the distinct relation PRESCRIPTION_ITEM rather than stored as comma-separated values in PRESCRIPTION.
    - Doctor schedule slots are maintained as individual tuples in DOCTOR_SCHEDULE.

⇒ Conclusion: All 13 relations satisfy 1NF.

2.3.3 Second Normal Form (2NF) Proof

Definition: A relation R is in 2NF if it is in 1NF and every non-prime attribute is fully functionally dependent on every candidate key of R (no partial functional dependencies).

Proof:

- Single-Attribute Candidate Keys:
    - The relations USER_ACCOUNT, PATIENT, DOCTOR, GENERAL_PRACTITIONER, SPECIALTY, DOCTOR_SCHEDULE, APPOINTMENT, CONSULTATION_SESSION, MEDICAL_HISTORY, PRESCRIPTION, and MEDICATION have candidate keys composed of a single attribute (e.g., UserID, PatientID, DoctorID, AppointmentID, SessionID).
    - Mathematical Property: For any single-attribute key K, no proper subset $K' \subset K$ exists ($K' \neq \varnothing$). Thus, partial dependency is mathematically impossible.
- Composite Candidate Keys — PRESCRIPTION_ITEM:
    - Consider candidate key K={PrescriptionID,MedicationID}.

- Non-prime attributes: Dosage, Frequency, Duration, SpecialInstructions.
- Neither PrescriptionID → Dosage nor MedicationID → Dosage holds individually because dosage is specific to a particular medication within a specific prescription.
- Therefore, non-prime attributes depend on the complete composite key{PrescriptionID,MedicationID}.

⇒ Conclusion: All relations satisfy 2NF.

2.3.4 Third Normal Form (3NF) Proof

Definition: A relation R is in 3NF if it is in 2NF and for every non-trivial functional dependency X → Y in F+, at least one of the following conditions holds:

- X is a superkey of R.
- Y is a prime attribute of R.

Proof:

- In APPOINTMENT: FD1: AppointmentID → All Attributes. X={AppointmentID} is a Superkey. No transitive dependencies exist among non-prime attributes ( example: PatientID ↛ Status ).
- In CONSULTATION_SESSION: FD1: SessionID → All Attributes and FD2: AppointmentID → All Attributes. Both X={SessionID} and X={AppointmentID} are Superkeys.
  - Note on Derived Attributes: ActualDurationMinutes is calculated from EndTime-StartTime. It is retained in the physical design strictly for query optimization, justified under performance guidelines.
- In DOCTOR: All determinant attributes (DoctorID, LicenseNumber, UserID) are candidate keys / superkeys.

⇒ Conclusion: No transitive dependencies exist. All relations satisfy 3NF.

2.3.5 Boyce-Codd Normal Form (BCNF) Proof

Definition: A relation $R$ is in BCNF if for every non-trivial functional dependency X → Y in $F^+$, X is a Superkey of R.

| **Relation** | **Functional Dependency (X→Y)** | **Is X a Superkey?** | **BCNF Status** |
|---|---|---|---|
| USER_ACCOUNT | UserID → All Attributes | YES ( UserID is PK) | PASSED |
| USER_ACCOUNT | Username → All Attributes | YES ( Username is Unique Key) | PASSED |

| | | | |
|---|---|---|---|
| PATIENT | PatientID → All Attributes | YES ( PatientID is PK) | PASSED |
| PATIENT | UserID → All Attributes | YES ( UserID is Unique FK) | PASSED |
| DOCTOR | DoctorID → All Attributes | YES ( DoctorID is PK) | PASSED |
| DOCTOR | LicenseNumber → All Attributes | YES ( LicenseNumber is Unique Key) | PASSED |
| DOCTOR_SCHEDULE | ScheduleID → All Attributes | YES ( ScheduleID is PK) | PASSED |
| APPOINTMENT | AppointmentID → All Attributes | YES ( AppointmentID is PK) | PASSED |
| CONSULTATION_SESSION | SessionID → All Attributes | YES ( SessionID is PK) | PASSED |
| CONSULTATION_SESSION | AppointmentID → All Attributes | YES ( AppointmentID is Unique FK) | PASSED |
| PRESCRIPTION_ITEM | PrescriptionItemID → All Attributes | YES ( PrescriptionItemID is PK) | PASSED |

| | | | |
|---|---|---|---|
| PRESCRIPTION_ITEM | {PrescriptionID, MedicationID} → All Attributes | YES ({PrescriptionID, MedicationID} is Candidate Key) | PASSED |

⇒ Final Conclusion: The MediConnect database schema strictly achieves Boyce-Codd Normal Form (BCNF).

## 3. Data Dictionary (Adapted from ISO/IEC 11179)

- This section specifies attribute details, data types, constraints, and business descriptions for all 13 relations in accordance with ISO/IEC 11179 metadata standards.

### 3.1 Table: USER_ACCOUNT

| Attribute Name | Data Type | Constraints | Description |
|---|---|---|---|
| UserID | INT | PK, Auto Increment | Unique surrogate identifier for user accounts |
| Username | VARCHAR(50) | UNIQUE, NOT NULL | System login credential |
| PasswordHash | VARCHAR(255) | NOT NULL | Encrypted password hash (SHA-256 / Bcrypt) |
| Role | VARCHAR(20) | CHECK (ADMIN, DOCTOR, PATIENT) | Role-Based Access Control role designation |

### 3.2 Table: PATIENT

| Attribute Name | Data Type | Constraints | Description |
|---|---|---|---|
| PatientID | INT | PK, Auto Increment | Unique surrogate identifier for patients |

| | | | |
|---|---|---|---|
| UserID | INT | FK, UNIQUE, NOT NULL | References USER_ACCOUNT.UserID |
| FullName | VARCHAR(100) | NOT NULL | Patient's full legal name |
| DateOfBirth | DATE | NOT NULL | Patient's date of birth |
| Gender | VARCHAR(10) | NOT NULL | Biological gender |
| Phone | VARCHAR(20) | NOT NULL | Primary contact telephone number |
| Email | VARCHAR(100) | NOT NULL | Electronic mail address |
| EmergencyContact | VARCHAR(100) | Nullable | Emergency contact person name and phone |

3.3 Table: DOCTOR

| **Attribute Name** | **Data Type** | **Constraints** | **Description** |
|---|---|---|---|
| DoctorID | INT | PK, Auto Increment | Unique surrogate identifier for medical doctors |
| UserID | INT | FK, UNIQUE, NOT NULL | References USER_ACCOUNT.UserID |
| FullName | VARCHAR(100) | NOT NULL | Doctor's full professional name |

| | | | |
|---|---|---|---|
| LicenseNumber | VARCHAR(50) | UNIQUE, NOT NULL | State medical board license number |
| Phone | VARCHAR(20) | NOT NULL | Office contact phone number |
| Email | VARCHAR(100) | NOT NULL | Professional email address |

3.4 Table: GENERAL_PRACTITIONER

| Attribute Name | Data Type | Constraints | Description |
|---|---|---|---|
| DoctorID | INT | PK, FK | References DOCTOR.DoctorID (EER Subtype) |

3.5 Table: SPECIALIST

| Attribute Name | Data Type | Constraints | Description |
|---|---|---|---|
| DoctorID | INT | PK, FK | References DOCTOR.DoctorID (EER Subtype) |
| SpecialtyID | INT | FK, NOT NULL | References SPECIALTY.SpecialtyID |

3.6 Table: SPECIALTY

| Attribute Name | Data Type | Constraints | Description |
|---|---|---|---|
| SpecialtyID | INT | PK, Auto Increment | Unique identifier for clinical specialty |

| | | | |
|---|---|---|---|
| SpecialtyName | VARCHAR(100) | UNIQUE, NOT NULL | Clinical specialty name (example: Cardiology,...) |

3.7 Table: DOCTOR_SCHEDULE

| Attribute Name | Data Type | Constraints | Description |
|---|---|---|---|
| ScheduleID | INT | PK, Auto Increment | Unique schedule slot identifier |
| DoctorID | INT | FK, NOT NULL | Doctor assigning availability |
| ScheduleDate | DATE | NOT NULL | Calendar date of working shift |
| StartTime | TIMESTAMP | NOT NULL | Shift start time |
| EndTime | TIMESTAMP | NOT NULL, CHECK (EndTime > StartTime) | Shift end time |
| AvailabilityStatus | VARCHAR(20) | CHECK (Available, Unavailable, Booked) | Slot reservation status |

3.8 Table: APPOINTMENT

| Attribute Name | Data Type | Constraints | Description |
|---|---|---|---|
| AppointmentID | INT | PK, Auto Increment | Unique appointment record identifier |

| | | | |
|---|---|---|---|
| PatientID | INT | FK, NOT NULL | Booking patient |
| DoctorID | INT | FK, NOT NULL | Assigned attending doctor |
| ScheduleID | INT | FK, NOT NULL | Associated schedule slot |
| FollowUpAppointmentID | INT | FK, Nullable | Self-referencing FK for follow-up chain |
| AppointmentDateTime | TIMESTAMP | NOT NULL | Scheduled date and time |
| EstimatedDurationMinutes | INT | NOT NULL | Estimated duration in minutes |
| AppointmentType | VARCHAR(20) | CHECK (Inperson, Telemedicine) | Consultation mode |
| Status | VARCHAR(20) | CHECK (Scheduled, CheckedIn, Completed, Cancelled, NoShow) | Appointment lifecycle status |

3.9 Table: CONSULTATION_SESSION

| **Attribute Name** | **Data Type** | **Constraints** | **Description** |
|---|---|---|---|

| | | | |
|---|---|---|---|
| SessionID | INT | PK, Auto Increment | Unique clinical session identifier |
| AppointmentID | INT | FK, UNIQUE, NOT NULL | Originating appointment |
| DoctorID | INT | FK, NOT NULL | Attending physician |
| PatientID | INT | FK, NOT NULL | Attending patient |
| StartTime | TIMESTAMP | NOT NULL | Actual session start timestamp |
| EndTime | TIMESTAMP | NOT NULL | Actual session end timestamp |
| ActualDurationMinutes | INT | CHECK (> 0) | Computed actual session duration |
| SessionType | VARCHAR(20) | CHECK (Inperson, Virtual) | Actual consultation setting |
| MeetingReference | VARCHAR(255) | Nullable | Video conference URL (for Virtual mode) |
| DiagnosisNotes | TEXT | Nullable | Physician clinical notes |

3.10 Table: MEDICAL_HISTORY

| Attribute Name | Data Type | Constraints | Description |
|---|---|---|---|
| MedicalHistoryID | INT | PK, Auto Increment | Unique medical record identifier |
| PatientID | INT | FK, NOT NULL | Owner patient |
| AppointmentID | INT | FK, NOT NULL | Associated appointment |
| SessionID | INT | FK, NOT NULL | Generating consultation session |
| RecordDate | TIMESTAMP | NOT NULL | Record creation timestamp |
| Diagnosis | TEXT | NOT NULL | Primary medical diagnosis |
| Symptoms | TEXT | Nullable | Observed patient symptoms |
| ProgressNotes | TEXT | Nullable | Longitudinal progress notes |

3.11 Table: PRESCRIPTION

| Attribute Name | Data Type | Constraints | Description |
|---|---|---|---|
| PrescriptionID | INT | PK, Auto Increment | Unique prescription header identifier |
| PatientID | INT | FK, NOT NULL | Recipient patient |

| | | | |
|---|---|---|---|
| DoctorID | INT | FK, NOT NULL | Prescribing physician |
| AppointmentID | INT | FK, NOT NULL | Associated appointment |
| SessionID | INT | FK, NOT NULL | Issuing consultation session. |
| DiagnosisICD | VARCHAR(50) | Nullable | International ICD diagnosis code |
| PrescriptionDate | TIMESTAMP | NOT NULL | Prescription issuance timestamp |
| Instructions | TEXT | Nullable | General prescription instructions |

3.12 Table: MEDICATION

| **Attribute Name** | **Data Type** | **Constraints** | **Description** |
|---|---|---|---|
| MedicationID | INT | PK, Auto Increment | Unique medication catalog identifier |
| MedicationName | VARCHAR(100) | NOT NULL | Generic or brand pharmaceutical name |
| Description | TEXT | Nullable | Pharmaceutical description and warnings |

3.13 Table: PRESCRIPTION_ITEM

| **Attribute Name** | **Data Type** | **Constraints** | **Description** |
|---|---|---|---|

| | | | |
|---|---|---|---|
| PrescriptionItemID | INT | PK, Auto Increment | Unique prescription line item identifier |
| PrescriptionID | INT | FK, NOT NULL | Parent prescription |
| MedicationID | INT | FK, NOT NULL | Prescribed medication item |
| Dosage | VARCHAR(100) | NOT NULL | Prescribed dosage (example: 500mg,...) |
| Frequency | VARCHAR(100) | NOT NULL | Administration frequency (example: Twice daily,...) |
| Duration | VARCHAR(100) | NOT NULL | Duration of therapy (example: 7 days,...) |
| SpecialInstructions | TEXT | Nullable | Special administration instructions |

4. Database Implementation (SQL Style Guide Compliant)

4.1 DDL Script (Tables, Views, Indexes, Triggers)

4.2 Advanced Queries & Performance Test Cases

5. Verification & Security

5.1 Test cases showing constraints prevent bad data insertion

5.2 Role-Based Access Control (RBAC) definition (GRANT/REVOKE statements)